Debre Markos University

College of Natural and Computational sciences

Department of Mathematics

Gradient Boosting Approach in Sales Prediction

By: Zegeju Tenaw

Advisor: Getachew Alemu (PhD)

January 2025

Debre-Markos, Ethiopia

Gradient Boosting Approach in Sales prediction

A Thesis Submitted to Debre-Markos University College of Natural and Computational Sciences Department of Mathematics for Partial fulfillment of the Requirement For the award of the degree of Master of Science in Mathematics (Algebra).

By

Zegeju Tenaw

Advisor: Getachew Alemu (PhD)

January, 2025

Debre-Markos, Ethiopia

DEBRE-MARKOS UNIVERSITY

DEPARTMENT OF MATHEMATICS

APPROVAL OF THE BOARD OF EXAMINERS

Approval sheet

The undersigned hereby certify that they have read and recommend to the College of Graduate Studies for acceptance a thesis entitled "Gradient Boosting Approach In Sales Prediction" by Zegeju Tenaw in partial fulfillment of the requirements for the award of the degree of Master of Science in Mathematics.

Advisor: Getachew Alemu (PhD): ————

External Examiner (1): ————————

External Examiner (2): ————————

Internal Examiner: ————————

Declaration

I hereby declare that the thesis entitled "Gradient Boosting Approach In Sales prediction" has been carried out originally by me, under the guidance and supervision of Dr. Getachew Alemu , Debre Markos University, College of Science, for the award of the degree of Masters in Mathematics and that this work has not been submitted either in whole or in part for any degree, diploma, or fellowship at any University.

Name: Zegeju Tenaw

Signature: ....., Date: .....

Place and date of submission: Debre-Markos University, Date: January 2025

With my approval as University Advisor, this thesis has been submitted for Examiner.

Getachew Alemu (PhD)

Signature ..... Date: .....

Contents

Acknowledgement v

Abstract vi

1. Introduction 1

1.1Background of the study 1

1.2 Motivation 3

1.3 Objectives Of the study 3

Thesis structure 4

2. Related work 5

3. Preliminaries 8

3.1 Graphs 8

3.2 Trees 9

3.3 Machine learning 10

4. Gradient boosting decision tree algorithm (GBDTA) 10

4.1 Overview 10

4.2 Structure of GBDT 14

4.3 Mathematical structure of GBDTA 17

5. GBDTA in sales prediction 23

5.1 Dataset description 23

5.2 Data cleaning and processing 25

5.2.1 Feature scaling 26

5.3 Data Analysis 26

6. Conclusion 31

Future Work 32

REFERENCES 33

Acknowledgement

First and foremost, I would like to express my thanks to my almighty God, who helped me from the beginning to the end of my thesis work. Next to this, I would like to express my deepest spatial thanks of gratitude to my advisor Getachew Alemu (PhD), who helped me to do this thesis and for his intellectual guidance, foremost encouraging and giving critical comments as well as correcting my errors by devoting a lot of time from the beginning to the end of my thesis work. After that, I would like to thank all the teachers who taught me. I'd also like to thank all of my friends for their valuable technical assistance during the stage of my thesis work. Thanks to my family e specially my husband Zelalem Tefera for their appreciation of my ability and he covers for any payments to accomplish my thesis work. And, I would like to thank all my teachers and the Department of Mathematics.

.

Abstract

Machine learning is transforming every part of life and has a major contributor in real world scenarios. The revolutionary applications of Machine learning can be seen in every field including education, healthcare, engineering, sales, entertainment, transport and several more. The traditional approach of the sales and marketing goals no longer help the companies to cope up with the pace of competitive market, as they are carried out with no insights to customers purchasing patterns. Major transformations can be seen in the domain of sales and marketing as a result of Machine learning advancements. The aim of this thesis is to propose a model for predicting the future sales of car seller companies keeping in view the sales of previous years.

The research involves data collection, preprocessing, feature selection, and model training, with performance evaluated using metrics such as Mean Squared Error (MSE) and R^2 score to assess accuracy and effectiveness. The key contributions of this thesis lie in its application of machine learning to predict future sales, thereby providing valuable insights for businesses to enhance their operations and strategic planning.

Key terms: Machine learning, GBDTA, Sales prediction, and Decision trees

1. Introduction

1.1Background of the study

Sales means you will give something when you gate many or revenue. It reacts of sailing a product or goods and the transaction between the sealer and the buyer.

Sales prediction also known as sales forecasting is the process of estimating a company's expected sales revenue over a specific time. Sales forecasting is focused on predicting future sales of a product or service with the help of historical data, market trends and other relevant factors [1, 2]. Sales forecasting adds excellent value to the business because it deals with the future. The primary goal of sales forecasting is to enable better financial planning, budget allocation and strategic decision making [3, 4] .

Targeting the market audience is the major focus of business sectors. It is therefore important that the company has been able to achieve the objective by employing a system of forecasting. The process of forecasting involves analyzing data from various sources such as market trends, consumer behavior and other factors [5]. This analysis would also help the companies to manage the financial resources effectively. The forecasting process can be used for many purposes, including: predicting the future demand of the products or service, predicting how much of the product will be sold in a given period. Sales forecasting methods are also in full bloom, such as time series model, machine learning model and deep neural network model [6].

For this thesis we apply the machine learning model to forecast future sales of a car seller company.

Machine Learning is field of study that gives computers the ability to learn without being explicitly programmed [7].

Machine Learning is the study of algorithms that improve their performance P at some task T with experience E: Definition by Tom Mitchell (1998): A well-defined learning task is given by <P,T,E>.

Machine learning is transforming every part of life and has a major contributor in real world scenarios. The traditional approach of sales and marketing goals no longer help the companies to cope up with the pace of the competitive market, as they are carried out with no insights to customers purchasing patterns. Major transformations can be seen in the domain of sales and marketing as a result of Machine learning advancements [8].

In recent years, the significance of accurate sales prediction has become increasingly apparent, particularly in the context of businesses striving to optimize their operations and enhance profitability. This thesis delves into the application of machine learning algorithms, specifically the Gradient Boosting Decision Tree Algorithms (GBDTA), to forecast sales effectively.

In this paper we have proposed the machine learning algorithms towards the data collected from the previous sales of a car seller company giving correlation, R^2 score and mean squared error to evaluate the prediction results of these models. The main advantages of this thesis are to help in predicting sales and improve the business by giving the ideas of future sales.

Gradient boosting is a technique used in creating models for prediction [7]. The technique is mostly used in regression and classification procedures. Prediction models are often presented as decision trees for choosing the best prediction. It works with three basic elements, such that loss function, weak learners to predict and a composite model that adds weak learners in a bid to improve the target outcome and minimize the prediction error.

1.2 Motivation

The reason that I am going to do on this title is it based on the literature, many researchers can forecasted future sales using a group of algorithms with GBDTA and they camper their performance and they select the best machine learning algorithm. For this thesis we select only the gradient boosting decision tree algorithm (GBDTA) and we goat good performance accuracy.

The motivation behind this research stems from the need for reliable sales forecasting in Ethiopia, where many companies operate without robust predictive frameworks. By leveraging machine learning algorithms, businesses can gain insights into future sales trends, thereby enhancing their operational efficiency and competitiveness.

1.3 Objectives Of the study

The general objective of the thesis is to predict the future sales of a car seller company by utilizing machine learning algorithms based on historical sales data. This goal is crucial for businesses that rely on accurate sales forecasts to drive their operational decision-making.

Additionally, the specific objectives of the study include several key aspects. First, it aims to develop insights into future sales trends, enabling businesses to strategize effectively and enhance their growth. Second, the research focuses on determining the amount of product that will be required in the future, which is essential for inventory management and ensuring that supply meets expected demand.

Moreover, the study seeks to analyze past sales data in order to inform and guide business decisions and strategies. By understanding historical sales patterns and consumer behavior, companies can improve their planning processes, thus maximizing their efficiency and competitiveness in the market. Overall, the objectives of this study highlight the importance of leveraging advanced analytical techniques for forecasting purposes in the sales domain.

Thesis structure

The thesis is structured into six main chapters, each encompassing various sections that detail the research and findings.

The first chapter, titled "Introduction," presents fundamental definitions of sales prediction and an overview of the role of machine learning algorithms in this domain. It further elaborates on the motivation behind the study and provides a general perspective on machine learning and its type.

The second chapter, "Related Work," offers a literature review that includes an analysis of previous studies relevant to the topic. This section examines different algorithms that have been used for sales prediction across various sectors, such as supermarkets and retail companies, while also discussing the performance metrics of these algorithms.

In the third chapter, the thesis introduces preliminary concepts such as graphs and trees before delving into the Gradient Boosting Decision Tree Algorithm (GBDTA).

The fourth chapter, this section describes the gradient boosting decision tree algorithm in detail, the overall structure of gradient boosting decision tree algorithms and the general procedure of GBDTA.

The fifth chapter, titled ‘’Implementation and Results," provides a comprehensive study and analysis of the predictive models. It includes the implementation of the GBDTA and evaluates its performance using various metrics, such as correlation coefficients, R^2 scores, and mean squared errors.

Finally, the six chapters, "Conclusion," summarize the findings of the research and discuss potential future work that could stem from the study, thereby highlighting areas for further exploration in the field of sales prediction.

2. Related work

As sales is one of the most important factor or any retailor in industry so the forecasting of that sales and analyzing it has been done by many researchers which is summarized

In 2001, Brown and Smith conducted early research on sales forecasting methods, emphasizing the limitations of traditional statistical approaches. Their work laid the groundwork for exploring more advanced techniques, highlighting the need for improved accuracy in sales predictions

Gradient boosting is an ensemble learning technique that builds models sequentially, with each new model aiming to correct the errors made by previous ones. This approach has been shown to be particularly effective in handling structured data, making it a popular choice for sales prediction tasks. Researchers such as Friedman (2001) introduced the foundational algorithms of gradient boosting, highlighting its ability to minimize loss functions and improve model performance iteratively.

In 2010, Gupta and Singh focused on traditional forecasting methods, illustrating their limitations in coping with rapidly changing market dynamics. Their work emphasized the need for more sophisticated approaches to improve sales accuracy.

By 2015, Wang et al. introduced a hybrid model that combined time series analysis with machine learning techniques. Their research demonstrated enhanced predictive capabilities and laid the groundwork for integrating various algorithms in sales forecasting.

In 2016, Garcia et al. explored the integration of machine learning algorithms in retail sales forecasting. They demonstrated that models such as Random Forest and Gradient Boosting could significantly enhance prediction accuracy compared to conventional methods. Their findings underscored the potential of machine learning to transform sales forecasting practices.

Applications in Retail Sales Prediction

Several studies have successfully applied gradient boosting models in retail sales forecasting. For instance, Chen et al. (2016) utilized extreme gradient boosting (XGBoost), a scalable implementation of gradient boosting, to predict sales for a large retail chain. Their findings demonstrated significant improvements in prediction accuracy compared to traditional linear models. Similarly, Kourentzes et al. (2014) explored the effectiveness of gradient boosting in forecasting demand for various products, revealing that it outperformed other forecasting techniques, particularly in volatile market conditions.

In 2017, Chen and Zhao explored the application of deep learning models in sales predictions. Their findings underscored the potential of neural networks to analyze complex data sets and generate more accurate forecasts.

Continuing in this vein, in 2018, Kumar et al. examined the comparative performance of multiple machine learning algorithms in the retail sector. Their study showed that decision tree-based models, particularly Gradient Boosting, outperformed traditional methods in predicting future sales.

In 2018, Wang and Chen focused on the application of Gradient Boosting Decision Trees (GBDT) in the automotive industry, demonstrating its effectiveness in analyzing customer behavior and predicting sales trends. Their research highlighted GBDT's ability to uncover complex relationships within data that traditional forecasting methods often miss.

In 2019, Johnson and Lee conducted a comprehensive review of sales prediction models across various sectors, emphasizing the increasing significance of machine learning in retail. They found that models employing ensemble methods, such as Gradient Boosting, significantly outperformed linear regression models in terms of predictive accuracy.

In 2020, Smith et al. explored different machine learning techniques for sales forecasting, showcasing the effectiveness of decision trees in capturing intricate patterns in sales data. Their study revealed that traditional forecasting methods frequently lack the accuracy provided by advanced algorithms.

In 2023, Thompson and Nguyen conducted a study focusing on the latest advancements in machine learning for sales prediction. They highlighted the emergence of new algorithms and techniques, such as deep learning, which further enhance the accuracy and efficiency of sales forecasting models.

An accurate sales prediction model using ML and feature engineering is conducted by XieDairu et al also eXtreme Gradient Boosting (XGBoost) is used for utilizing these features to forecast the sales amount, various algorithms where compared and result concluded that xgboost performs better with RMSE 0.655 also it performed well with less computing time and for memory resources.

A grid search optimization technique was suggested by GopalBehera et al to optimizing the parameters and selecting the best tuning hyper parameter also used XG Boost (before

tuning parameter) and (After tuning parameter) hence the model is performing better when the hyper parameters are tuned.

During the experiment inYiyangNiu et al implemented detailed feature engineering procession such as memory compression, statistical features, temporal feature selection and it presents the feature engineering is more effective for training model.

SunithaCheriyan Presents an investigation of how decisions are made based upon data and insights that are gained by visualizing data, as well as the usage of data mining tools. The gradient boost method has been demonstrated to have the highest level of accuracy in predicting future transactions.

According to Walmart sales prediction (Best ML Algorithm), Walmart builds Regression models to predict the sales with respect to a single and multiple features and runs several promotional markdown events throughout the year. These markdowns precede **prominent** holidays, the four largest of all, which are the Super Bowl, Labor Day, Thanksgiving, and Christmas. The weeks including these holidays are weighted five times higher in the evaluation than non-holiday weeks. Part of the challenge presented by this competition is modeling the effects of markdowns on these holiday weeks in the absence of complete/ideal historical data.

While many studies explore various machine learning algorithms, this thesis specifically concentrates on the Gradient Boosting Decision Tree Algorithm (GBDTA). It provides a detailed can be effectively applied to sales prediction, highlighting its advantages over other algorithms.

3. Preliminaries

3.1 Graphs

Definition: A graph is a set of vertices and edges where each edge is connected between vertices and edges. In mathematically, A graph G=(V,E) is a set of vertices V and edges E where each edge (u,v) is a connected between vertices where u & v in V

There are three types of graph: Directed graph, undirected graph and mixed graph

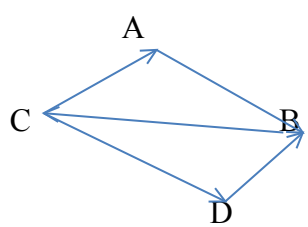
1. A directed graph (or digraph) is a type of graph where the edges have a direction. This means that each edge connects an ordered pair of vertices, indicating a one-way relationship between them. In a directed graph, if there is an edge from vertex A to vertex B, it does not imply that there is an edge from B to A.

Example:

Consider a directed graph with the following vertices: A, B, C, D, A.

- Edges:
    - A→B (A points to B)
    - B→C(B points to C)
    - C→ *A* (C points to A)
    - D→B (D points to B)

This directed graph can be visualized as:

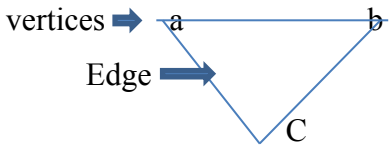


2. Undirected graph: An undirected graph is a type of graph in which the edges do not have a direction. This means that each edge connects two vertices (or nodes) and indicates a two-way relationship between them.

In an undirected graph:

- Each edge is represented as an unordered pair {u, v}, where u and v are the vertices connected by the edge.

The connection between the vertices is bidirectional, meaning that if there is an edge between u and v, you can traverse from u to v and from v to u without any restriction.



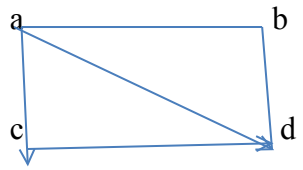
For example

V={a,b,c} and E={(a,b),(b,c),(a,c)}

path=a-b-c, a-c-b, c-b-a, b-c-a and c-a-b

3. Mixed graph: A mixed graph is a type of graph that contains both directed and undirected edges. In a mixed graph, some edges have a direction (indicating a one-way relationship between vertices), while others do not (indicating a two-way relationship).

For example



v={a,b,d,c} and E={(a,b),(b,d),(a,d),(a,c),(c,d)}

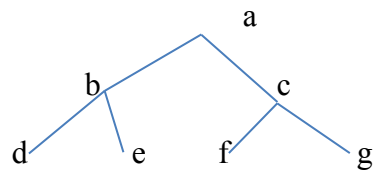
Path= (a, b, d), (a, c, d), (a, d) is called mixed graph

3.2 Trees

Definition: A Tree is an undirected graph that satisfies

1. T is connected and Acyclic
2. T is acyclic and a simple cycle is formed if any edge is added to G
3. Any two vertices in T can be connected by a simple unique path
4. T is connected and has n-1 edges

For example



This is a tree that has three internal nodes (vertices) and four external nodes (vertices).

Each internal vertex has a minimum of two edges incident to that vertex. Internal vertex has degree more than or equal to two but an external vertex has only one edge incident to that vertex it have only one degree. External vertices are called leaf nodes because they do not have extra edges.

A leaf is **a vertex with no children**. An internal vertex is a vertex that is not a leaf. The height of a vertex in a rooted tree is the length of the longest downward path to a leaf from that vertex. The height of the tree is the height of the root.

3.3 Machine learning

Machine learning is a branch of artificial intelligence that involves algorithms to make predictions or decisions based on data. In general, Machine Learning is a program that can manage various tasks by analyzing and exploring data [9, 10]

Types of Machine Learning

Supervised Learning:

The most popular model for performing Machine Learning processes is supervised learning. It is commonly used for data where the mapping between input-output data is accurate. Supervised learning uses labeled data to train machine learning models. Labeled data means that the outputs are already known to you. The model just maps the input to be the output. Supervised machine learning is used for this article to create machine learning models used for prediction and classification purposes .In general given: training data + desired outputs (labels)

Unsupervised Learning:

The data is not explicitly labeled into different classes in the case of unsupervised learning, which is there is only unlabeled data. By identifying implicit patterns the model can learn from the data [11]. Unlabeled data means there is no fixed output variable. That is training data without desired outputs.

Reinforcement Learning:

Reinforcement Learning is a sub-field of Machine Learning. In a given scenario, it is about taking appropriate action to optimize reward. Various algorithms and computers are employed to determine the best possible action or path it will follow in a specific scenario.

4. Gradient boosting decision tree algorithm (GBDTA)

We review the gradient boosting decision tree in this section. The derivation follows from the same idea in existing literature on gradient boosting. The second order method originated from Friedman et al [12]. We make minor improvements in the regularized objective, which were found helpful in practice.

4.1 Overview

Gradient boosting decision tree algorithms (GBDT) is a supervised machine learning algorithm that uses sequentially boosted trees. It uses the training data (with multiple features) x_i to predict a target variable y_i based on the features. Before we look at the structure of gradient boosting decision tree algorithms, let us start by reviewing the definition of boosting.

Boosting:

An ensemble technique that builds models sequentially. Each model learns from the mistakes of the previous model.

It works on the principle of improving mistakes of the previous learner through the next learner. In this learning technique, weak learners are used which perform only slightly better than a random chance.

Boosting focuses on sequentially adding up these weak learners and filtering out the observations that a learner gets correct at every step. Basically the stress is on developing new weak learners to handle the remaining difficult observations at each step. Gradient boosting decision tree algorithm works with the principle of improving prediction/classification results using sequential decision tree models.

Components of GBDT algorithm

- Leaf index mapping function
- Leaf weight
- Objective function
- Loss Function
- Regularize

Leaf index mapping function

In the context of decision trees and tree-based algorithms (like those used in scikit-learn), the **leaf index mapping function** refers to the process of mapping each data point to the corresponding leaf node in the tree where it ends up after being processed through the decision tree.

Here's a breakdown of its significance:

Leaf Nodes: In a decision tree, leaf nodes represent the final output of the decision process. Each leaf contains a prediction based on the features of the data points that reach it.

Mapping Function: The mapping function takes input features and traces them down the tree, determining the path based on the conditions at each node until it reaches a leaf. This process results in assigning an index (or identifier) to each leaf.

Usage: The leaf index can be useful for:

Feature Importance: Understanding which features contribute to predictions by analyzing the leaves where instances fall.

Ensemble Learning: In methods like Random Forests, the mapping can help in aggregating predictions from multiple trees.

Interpretability: It can aid in interpreting model predictions, showing how specific features influence outcomes.

Overall, leaf index mapping provides a structured way to track where each input data point is classified in the decision tree, facilitating better understanding and optimization of model predictions.

Leaf weight

In the context of decision trees and tree-based algorithms, **leaf weight** refers to the importance or contribution of a particular leaf node to the overall prediction made by the model. Here are key points about leaf weight:

Leaf weight is typically calculated as the number of samples (or data points) that reach that leaf node during the training phase. It represents how significant that leaf is in terms of the volume of data it handles.

Importance in Predictions: The weight of a leaf can influence the final predictions of the model. For example, in ensemble methods like Random Forests, predictions from different trees are often aggregated based on the weights of the leaves that the input samples reach.

Relation to Class Distribution: In classification problems, the leaf weight can also reflect the distribution of classes among the samples that reach that leaf. This can help in understanding how confident the model is about its predictions for that specific region of the feature space.

Use in Interpretability: Analyzing leaf weights can provide insights into the decision-making process of the model, helping to interpret how certain features contribute to different predictions.

Generally, leaf weight is a useful concept for understanding and interpreting the behavior of decision trees and related algorithms.

Objective function

An **objective function** is a mathematical function that defines the goal of an optimization problem. In machine learning and statistics, it quantifies the performance of a model by measuring how well it fits the training data. Here's a deeper look at its significance:

The objective function typically takes input parameters (like model weights) and outputs a single scalar value that indicates the quality of the solution.

Loss Functions

In supervised learning, the objective function is often a loss function that measures the difference between the predicted values and the actual target values. Common examples include Mean Squared Error (MSE) for regression and Cross-Entropy Loss for classification.

Utility Functions: In some contexts, it can represent a utility that needs to be maximized (e.g., in reinforcement learning).

Optimization: The goal is to find the parameters that minimize (or maximize) the objective function. Optimization algorithms like gradient descent or genetic algorithms are used to adjust the parameters iteratively to achieve this.

Evaluation Metric: The value of the objective function helps evaluate the effectiveness of different models or configurations, guiding the selection of the best-performing model.

Constraints: In constrained optimization problems, the objective function is subject to certain constraints that limit the feasible solutions.

In general, the objective function plays a central role in training machine learning models, guiding the learning process and determining how well a model performs on a given task.

Regularization

Regularization is a technique used in machine learning and statistics to prevent over fitting, which occurs when a model learns the noise in the training data rather than the underlying pattern. By adding a penalty for complexity to the loss function, regularization helps to keep the model simpler and more generalizable to unseen data. Here are the key points:

Purpose: Regularization aims to improve a model's ability to generalize by discouraging overly complex models that fit the training data too closely.

Overall, regularization is a crucial technique in building robust machine learning models that perform well on new, unseen data.

4.2 Structure of GBDT

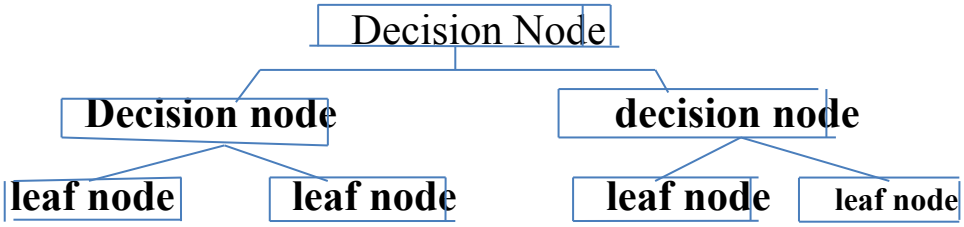
Gradient boosting decision trees use a sequence of steps. Here we explain the main steps of the algorithm.

Gradient boosting decision tree (GBDT) combines the strengths of multiple decision trees to create a robust predictive model. It iteratively builds trees, each one correcting the errors of the previous ones, resulting in a highly accurate ensemble. This chapter introduces the core principles of Gradient boosting decision tree (GBDT), exploring its relevance and superiority in sales prediction scenarios.

Gradient Boosting Decision Trees algorithm (GBDTA) is a machine learning technique that combines the predictions of multiple decision trees to improve accuracy. Gradient boosting decision tree (GBDT) models are good at capturing complex patterns and relationships in data. Using Gradient boosting decision tree (GBDT) for sales forecasting is a powerful approach due to its ability to handle complex, non-linear relationships in data. Here is an overview of how we can use GBDTA for sales forecasting. The main library in python computation helpful to apply Gradient boosting decision tree algorithms (GBDTA) is "sklearn".

Decision Trees: Simple models used as the building blocks.

Below diagram explains the general structure of a decision tree:



Note: A decision tree can contain categorical data (YES/NO) as well as numeric data.

Decision Tree Terminologies

Root Node: Root node is from where the decision tree starts. It represents the entire dataset, which further gets divided into two or more homogeneous sets.

Leaf Node: Leaf nodes are the final output node, and the tree cannot be segregated further after getting a leaf node.

Splitting: Splitting is the process of dividing the decision node/root node into sub nodes according to the given conditions.

Branch/Sub tree: A tree formed by splitting the tree.

Pruning: Pruning is the process of removing the unwanted branches from the tree.

Advantages of using GBDT

Adaptability: GBDT can handle various types of data, including numerical and categorical. It can also handle a mix of categorical and numerical features effectively.

Flexibility: Used in various applications, from classification to regression.

High Predictive Accuracy:

GBDT is known for high predictive accuracy

GBDT is one of the most accurate machine learning algorithms due to its ability to model complex, nonlinear relationships in the data. It often outperforms other models like decision trees and linear models.

Handles Outliers:

The method is less sensitive to outliers than many other algorithms, as it builds trees that focus on correcting the errors from the previous trees rather than being overly influenced by extreme data points.

Interpretability:

While individual trees in GBDT are relatively simple and interpretable, the overall ensemble model can be complex. However, techniques like **feature importance** (i.e., how much each feature contributes to the predictions) can help in understanding the model.

Robustness:

GBDT can handle various types of data, including missing values, and is generally robust to noise, as each tree focuses on correcting the previous errors.

In general GBDT becomes more reliable than the other regression in predicting probability for diabetes with big data.

Disadvantages of Gradient Boosting:

Over fitting:

If the number of trees is too large or the model is not regularized properly (e.g., not controlling tree depth or using an excessive learning rate), GBDT models can overfit to the training data, leading to poor generalization to unseen data.

Computational Cost:

Building trees sequentially makes GBDT computationally expensive, especially when using a large number of trees or deep trees. Training times can be long, particularly on large datasets.

General Procedures:

- **Initialize**: Start with an initial prediction, often the mean of the target variable for regression tasks.
- **Residual Computation**: Calculate the residuals, which are the differences between the actual values and the predicted values from the current model. These are how far off the model's predictions are from the actual values.
- **Train a New Decision Tree:** Fit a decision tree to predict these residuals. This tree becomes a weak learner that tries to correct the mistakes of the current model.

- **Update the Model:** Add the new tree's predictions to the existing model's predictions. This update is scaled by a learning rate (often denoted as ν\nu or η\eta) to control the influence of each new tree.
- **Repeat:** Repeat steps 2 to 4 for a predefined number of iterations or until the model reaches a certain performance threshold.
- **Combining the Models: T**he final prediction is the sum of the initial prediction and the weighted predictions from all the trees. This aggregation is done to form a strong overall prediction.

The key components of methodology for this research

- **Data Collection**: Collect the relevant data required for the analysis.
- **Data Preprocessing**

  Cleaning: Handle missing values, remove outliers.

  Feature Engineering: Create new features such as moving averages, lag features (e.g., sales from the previous day/week), and encode categorical variables like day of the week and weather conditions.
- **Splitting the Data**

Split the data into training and testing sets to evaluate the performance of the model. Typically, we might use data from the last few weeks or months as the test set.

- **Model Training**

Train the model on the training data. The model learns from historical patterns and relationships between features and sales. Each tree in the model helps to correct the errors of the previous trees, gradually improving the accuracy of the predictions.

- **Model Evaluation**

Evaluate the model on the test set using metrics like Mean Absolute Error (MAE) or Root Mean Squared Error (RMSE). This helps to understand how well the model generalizes to unseen data.

- **Forecasting**

Use the trained model to make future sales predictions. For example, we can predict sales for the next week or month, considering upcoming promotions and holidays.

4.3 Mathematical structure of GBDTA

Gradient Boosting Decision Trees (GBDT) follow a systematic sequence of mathematical computations designed to build a powerful predictive model by combining the strengths of multiple weak learners, typically decision trees [13]. This iterative approach ensures the model captures the underlying patterns in the data effectively. Below, we discuss the main mathematical computations involved behind the GBDT algorithm:

Gradient boosting decision tree uses a sequence of steps. Here we explain the main steps of the algorithm.

We initialize the algorithm by the dataset $\{(xi,yi)\}$i=1 to n and

a differentiable loss function is

L (yi, F (xi)) $=\frac{1}{2}\sum_{i=1}^{n}(yi - F(xi))2$

Step-1: Initializing the algorithm with constant value; single leaf tree.

y0^ =F0(x) =argmin $\sum_{i=1}^{n} L(yi,y)$

The first step gives a leaf that predicts all samples with the same value. This is the initial value of the algorithm. Moreover, since the loss function $L(y_i,\gamma)$ is the known mean square error (MSE), using the concept of calculus, we have $\hat{y}_0 = Average(y_1 + y_2 + ... + y_n)$. It is the initial prediction for all samples.

A single leaf tree that contains a prediction value F_0 for all the samples in the dataset.

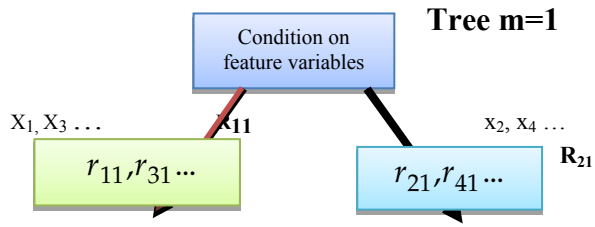
Step-2: Fix the number of tress: For $m = 1$ **to** M **(number of trees):**

a) Compute the pseudo residuals

$r_I = -$

for each sample and m^{th} – tree.

b) Create a terminal region R_{jm}, $j = 1,2,...j_m$ for j^{th} –leaf and m^{th} –tree that fits to each value of r_I for each samplex_i.



c) Compute output value for each leaf node

$$\gamma_{jm} = \underset{\gamma}{argmin} \sum_{x_i \in R_{ij}}^{n} L\left(y_i, F_{m-1}\left(x_i\right) + \gamma\right),$$

for $j = 1,2,...j_m$. For leaves containing more than two residuals, the output value can be obtained using the optimization problem over all samples in the region R_{ij} .

d) Make a new prediction

$F_m(x) = F_{m-1}(x) + \alpha \sum_{j=1}^{j_m} \gamma_{jm}$

For each sample by summing up all the output values γ_{jm} for a sample $x \in R_{jm}$. The learning rate α is always between 0 and1.

What is the implication for $\alpha = 0$ and $\alpha = 1$?

In the context of predictions, especially when discussing algorithms that involve regularization or weighted averages, the parameter α often controls the balance between different influences on the prediction. Here's what α=0 and α=1 typically imply:

1. **α=0**:
   - This value usually indicates that you are disregarding the influence of one component entirely.
   - In the context of regularization, setting α=0 would mean that no regularization is applied. The model would solely focus on minimizing the loss function without any penalty for complexity.
   - For weighted averages or mixing models,α=0 could imply that only one model or data source is being used, completely ignoring the others.

2. **α=1**:
    - Conversely, α=1 often indicates that you are fully relying on a specific component or model.
    - In regularization, setting α=1 may signify that you are applying full regularization, potentially leading to a model that is heavily biased but with reduced variance, as it avoids overfitting.
    - In a weighted scenario, α=1 might mean that you are fully incorporating the influence of one model while completely disregarding others.

Implications:

- **When α=0**:
    - The model may fit the training data very well (low bias) but could suffer from overfitting (high variance).
- **When α=1**:
    - The model may be overly simplistic, potentially under fitting (high bias) and not capturing the complexity of the data.

Through this iterative process, Gradient Boosting Decision Trees build a strong predictive model by progressively correcting the errors of the previous iterations. The careful combination of multiple weak learners ensures that the model captures the intricacies of the data, leading to high accuracy and reliable predictions.

Example

Here we illustrate the steps of the GBDTA using a tabular dataset. Here's how we can visualize.

Week	**Advertising Spend($)**	**Promotion Week**	**Sales**
1	**1000**	**No**	**200**
2	**1500**	**Yes**	**300**
3	**1200**	**No**	**250**
4	**1800**	**Yes**	**400**
5	**1700**	**No**	**350**

Step 1: Initial Prediction

- Initial Prediction for Sales: 300
- Mean Sales: Y=300

Step 2: Compute Residuals

- Residuals Calculation:
    - Week 1: 200-300=-100
    - Week 2: 300-300=0
    - Week 3: 250-300=-50
    - Week 4: 400-300=100
    - Week 5: 350-300=50

Step 3: Train First Decision Tree

Train a decision tree to predict residuals:

- Node 1: If Promotion Week = No→Predict -50
- Node 2 : If Promotion Week = Yes →Predict 50

Step 4: Update Model with Learning Rate α =0.1

Updated Predictions:

- Week 1: 300+ 0.1(-100) =290
- Week 2: 300+0.1(0) = 300
- Week 3 : 300+ 0.1(-50)=295
- Week 4: 300+0.1(100) = 310
- Week 5: 300+0.1(50) = 305

Step 5: Compute New Residuals

- Week 1: 200-290 =-90
- Week 2: 300-300 =0
- Week 3: 250-295 =-45
- Week 4: 400-310 =90
- Week 5: 350-305 =45

Step 6: Train Second Decision Tree

Train another decision tree to predict new residuals:

- Node 1 : If Promotion Week = No→Predict -45
- Node 2 : If Promotion Week = Yes→Predict 45

Step 7: Update Model with Learning Rate α =0.1

Updated Predictions:

- Week 1 : 290+0.1(-90)=281
- Week 2 : 300+0.1(0) = 300
- Week 3 : 295 + 0.1(-45) =290.5
- Week 4 : 310 + 0.1(90) =319
- Week 5 : 305 +0.1(45) =309.5

Step 8: Compute New Residuals Again

- Week 1: 200-281=-81
- Week 2 : 300-300 =0
- Week 3 : 250-290.5 =-40.5
- Week 4 : 400-319 =81
- Week 5 : 350-309.5 = 40.5

Through these iterations, GBDT continues to train new trees to predict the residuals, updating the model's predictions step by step. Each tree corrects errors from the previous trees, improving accuracy. As we see from the iterations the first prediction (assigning a constant value for each value) was weaker than the others. Moreover, the i-th iteration was built on the residuals of the (i-1)-th iteration.

In general, considering the residuals to build the next iteration in GBDT is crucial because residuals represent the errors made by the current model. By focusing on these residuals, the algorithm aims to minimize these errors and improve the model's predictions. Essentially, each new tree is trained to correct the mistakes of the previous trees. The main reasons for this are:

- **Focus on Errors**: By fitting a new tree to the residuals, the model specifically targets and reduces the remaining errors, making the overall prediction more accurate.
- **Sequential Learning**: This iterative process ensures that the model learns from its mistakes and refines its predictions step-by-step.

- **Optimal Correction**: By continuously adjusting for the residuals, the model incrementally approaches the best possible fit to the data.

This strategy allows GBDT to build a powerful model that leverages the strengths of multiple weak learners, correcting errors along the way. It is like fine-tuning a recipe until it is just right.

5. GBDTA in sales prediction

In the ever-evolving landscape of business, accurate sales forecasting is essential for making informed decisions and staying competitive. Traditional methods often fall short in capturing complex patterns and non-linear relationships within data. This is where Gradient Boosting Decision Trees (GBDT) comes into play—a powerful machine learning technique that has revolutionized the way we predict sales.

In this chapter, we apply the GBDT algorithm to predict the sales of a car seller company. We begin by describing the dataset obtained from the Kaggle website, which includes critical features such as customer name, customer email, country, gender, age, annual salary, credit card debt, net worth, and car purchase amount. This data provides valuable insights into customer demographics, income, and financial health.

We then delve into the application of the GBDT algorithm on this dataset. By leveraging the rich information contained within the dataset, the GBDT algorithm allows us to uncover complex patterns and relationships, providing a comprehensive view of customer behavior and sales trends.

This process highlights why GBDT is a game-changer for businesses aiming to forecast sales with precision and accuracy. With the availability of comprehensive data, GBDT can significantly enhance sales prediction, enabling businesses to make data-driven decisions and stay ahead in the competitive market.

5.1 Dataset description

The datasets obtained from the Kaggle website are from a car seller company, which encompass a range of critical information to ensure efficient and effective sales operations. These datasets include several important features, such as:

- **Customer Name**: The full names of the customers.
- **Customer Email**: Email addresses for communication and marketing purposes.
- **Country**: The country of residence of the customers.
- **Gender**: The gender of the customers.
- **Age**: The age of the customers.
- **Annual Salary**: The yearly income of the customers.

- **Credit Card Debt**: The amount of debt customers have accumulated on their credit cards.
- **Net Worth**: The total assets minus liabilities, indicating the financial health of the customers.
- **Car Purchase Amount**: The amount of money customers have spent on purchasing cars.

This data provides valuable insights into various aspects of customer demographics, such as their income, geographical location, gender, and age. Additionally, it offers information on financial attributes like annual salary, credit card debt, and net worth, which can be crucial for understanding customer purchasing power and behavior.

Collectively, these datasets offer a comprehensive view of customer information, which can be leveraged for future sales prediction. By analyzing this data, the car seller company can identify trends, preferences, and potential market segments, thereby making informed decisions to optimize their sales strategies and improve customer satisfaction.

Dimension:

The variables listed above are records of the seller during the transaction. Here we drop the features 'customer name', and 'customer email', 'country". In our analysis we have a dataset dimension of (6, 500). In machine learning (ML), **features** and the **target variable** are essential components to describe the dataset and the sales prediction problem. Let us break down the dataset in two. We give 80% of the dataset are for the training data set and the rest 20% of the dataset are for the testing dataset.

The first five rows of the dataset are shown bellow

Table-1

	gender	age	annual Salary	credit card debt	net worth	car purchase amount
0	0	41.851720	62812.09301	11609.380910	238961.2505	35321.45877
1	0	40.870623	66646.89292	9572.957136	530973.9078	45115.52566
2	1	43.152897	53798.55112	11160.355060	638467.1773	42925.70921
3	1	58.271369	79370.03798	14426.164850	548599.0524	67422.36313
4	1	57.313749	59729.15130	5358.712177	560304.0671	55915.46248

Features (X): Features are the input variables (represented as vectors) that we use to make predictions. They represent various attributes or properties of the data. For us there are five feature variables: **'gender', 'age', 'annual Salary', 'credit card debt', 'net worth'.**

Target variable (y): The target variable (also known as the label) is what we are trying to predict. It is the output variable in our dataset. In a dataset, the target variable is often represented as one column. In this case, **"car purchase amount"** is the target variable.

5.2 Data cleaning and processing

Data cleaning is an essential step in the data analysis process to ensure the quality and accuracy of the data. It involves identifying and correcting (or removing) missing values and inconsistencies in the data to improve its reliability. There are different techniques of data cleaning; here we use the technique of scaling.

5.2.1 Feature scaling

The main purpose of scaling is to ensure that all features contribute equally to the model's predictions, avoiding biases that could occur due to varying feature values. Here we use the **MinMax scaler.**

It is a data preprocessing technique used in machine learning to normalize or scale data features using the formula:

$$X\ new = \frac{(X\ old - X\ \min\ )}{(X\ \max\ - Xmin)}$$

where

X_old is the original feature value.

X_min is the minimum value of the feature.

X_max is the maximum value of the feature.

X_new is the scaled feature value.

This ensures that the transformed values are scaled within the range the interval [0, 1].

5.3 Data Analysis

In this section, we analyze the dataset obtained from the car seller company to understand the underlying patterns and relationships that influence sales. The dataset includes features such as gender, age, annual salary, credit card debt, net worth, and car purchase amount. We have done the data cleaning, handling missing values, and standardizing formats. We visualize the distribution of key features and identify correlations between them. This analysis reveals insights into customer demographics, financial status, and purchasing behavior, setting the stage for applying the Gradient Boosting Decision Trees (GBDT) algorithm. By understanding these relationships, we can effectively train the GBDT model to make accurate sales predictions, ultimately enhancing the company's decision-making process and sales strategies.

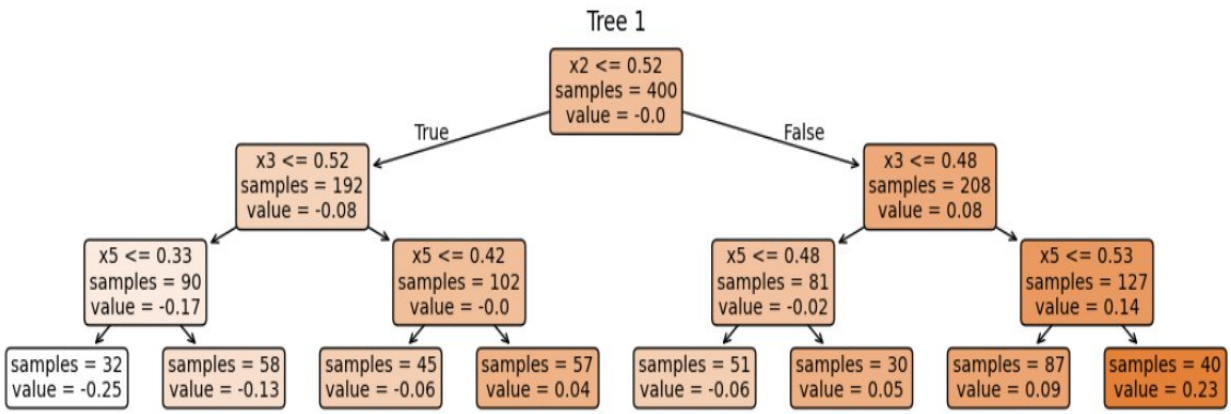
Instead of manually calculating our weak learners, we are going to use the builtin function **Gradient Boosting Regressor**, which is what, happens in the real world.

Sample trees in the GBDT algorithm:

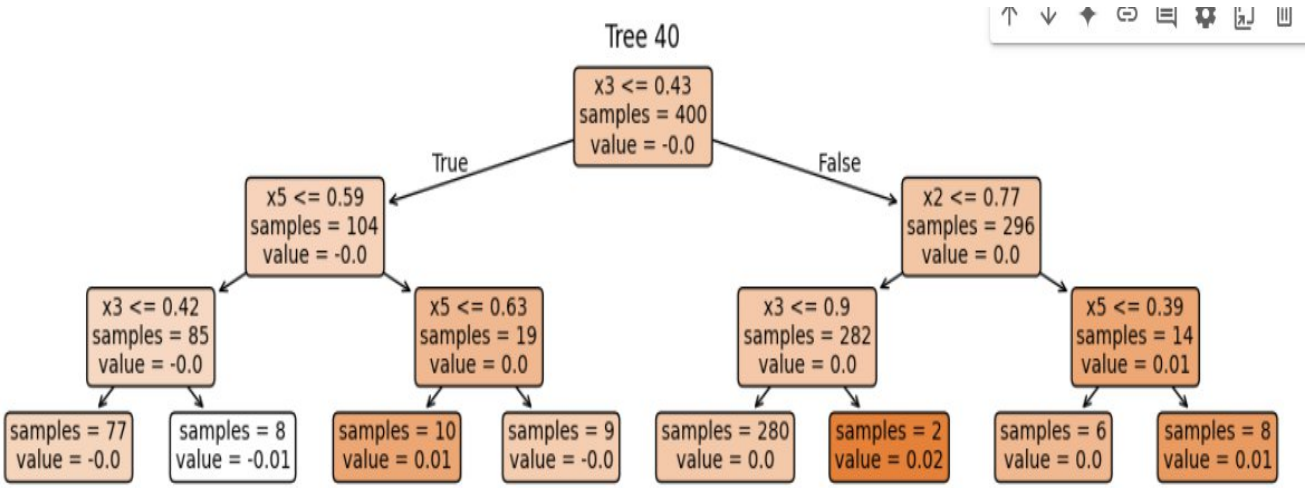
In our Gradient Boosting Decision Trees (GBDT) car sales prediction model, individual decision trees play a crucial role in refining predictions. The model begins with an initial prediction. Each subsequent tree is built to correct the residuals (errors) from previous predictions. Over multiple

iterations, these trees collectively enhance the model's accuracy by iteratively reducing the error and converging towards a more precise prediction of sales.

Here we give the structure of the first tree .



Moreover, the structure of the final regression tree is given as



This tree starts giving a condition on the third feature variable.

RMSE (Root Mean Squared Error) and R^2 score:

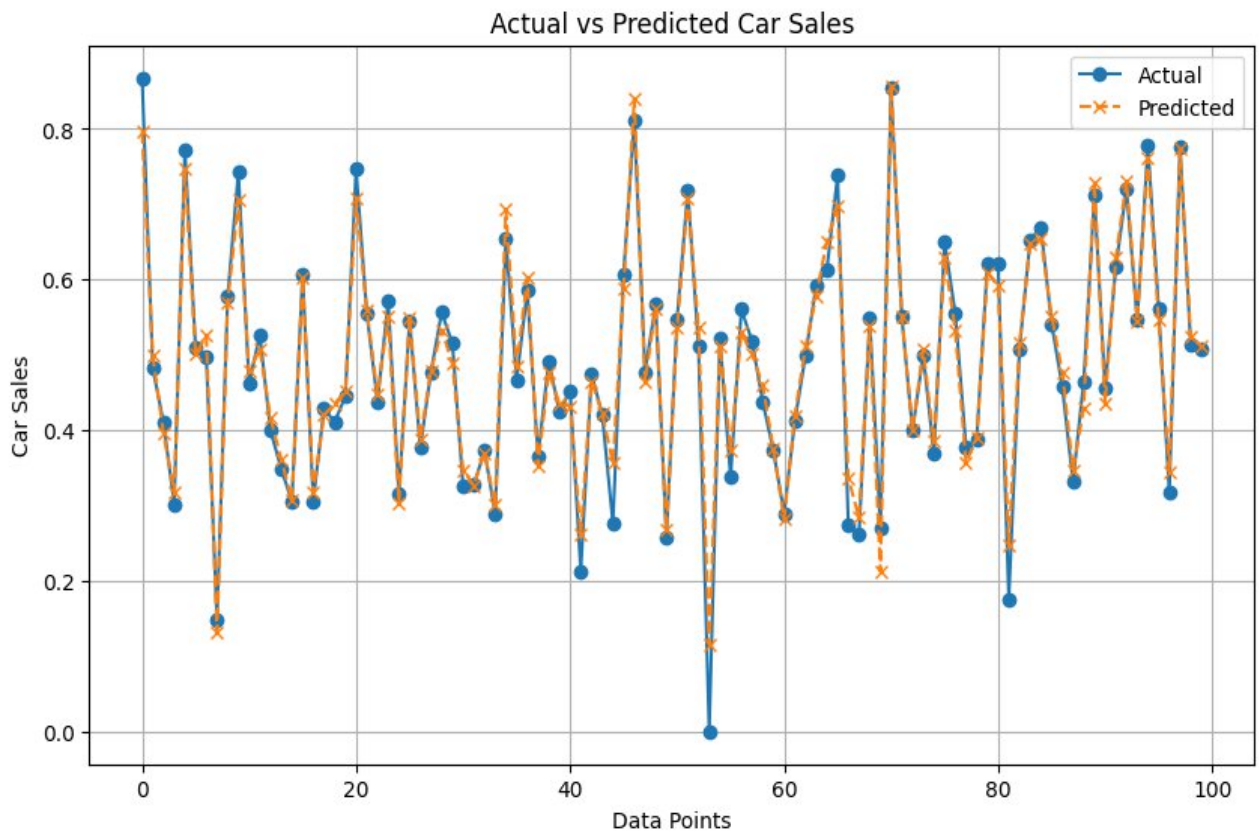
The predictive model for car sales demonstrates exceptional performance, as evidenced by the key metrics of RMSE (Root Mean Squared Error) and R^2 score. The model's RMSE of 0.0262 indicates that the average difference between the predicted car sales and the actual sales figures is remarkably low. This minimal error margin underscores the model's precision in forecasting car sales, ensuring that the predictions closely align with the actual sales data.

	Actual	**Predictive**
0	0.867591	0.796201
1	0.482290	0.498835
2	0.410402	0.395806
3	0.301687	0.316751
4	0.772802	0.746591
..	…	…
95	0.561992	0.546114
96	0.316780	0.343822
97	0.776722	0.773396
98	0.513270	0.524547
99	0.508552	0.511163

[100 rows x2 columns]

Model RMSE:0.0264

Additionally, the R^2 score of 0.973 signifies that the model explains 97.3% of the variance in car sales. This high explanatory power highlights the model's ability to capture the underlying trends and patterns within the data, leaving only 2.7% of the variability unexplained. Collectively, these metrics reflect a robust and reliable prediction model that can significantly enhance the accuracy of sales forecasting for the car seller company. Such a model empowers the business to make well-informed decisions, optimize inventory management, and develop targeted marketing strategies, ultimately driving improved sales performance and customer satisfaction based on ‘age’, ‘ annual salary’, ’credit card debt’, ’net worth’, of the customers.



This includes the actual vs predicted sales graph, which typically provides a visual representation comparing the real sales data with the sales predictions made using the Gradient Boosting Decision Tree Algorithm (GBDTA).In such graphs:

- X-axis represents data points from the target variable
- Y-axis represents the predicted car sales

The graph would typically display:

- Actual Sales: This is plotted as a line or set of points representing the historical sales figures gathered from the car seller company.
- Predicted Sales: This would be another line or set of points that represent the sales figures forecasted by the machine learning model .The objective of this comparative graph is to visually assess how closely the predicted sales align with the actual sales. A well-fit model will show the predicted sales line closely following the actual sales line, indicating high accuracy in the forecasting.

Residual plot

The residual plot for the sales prediction model of cars shows a random scatter of residuals around the zero line without any discernible patterns. This indicates that the model's predictions are unbiased and that it captures the underlying patterns in the data well. The absence of systematic structures or trends in the residuals suggests that the model's assumptions are appropriate and that it performs consistently across different ranges of predicted values.

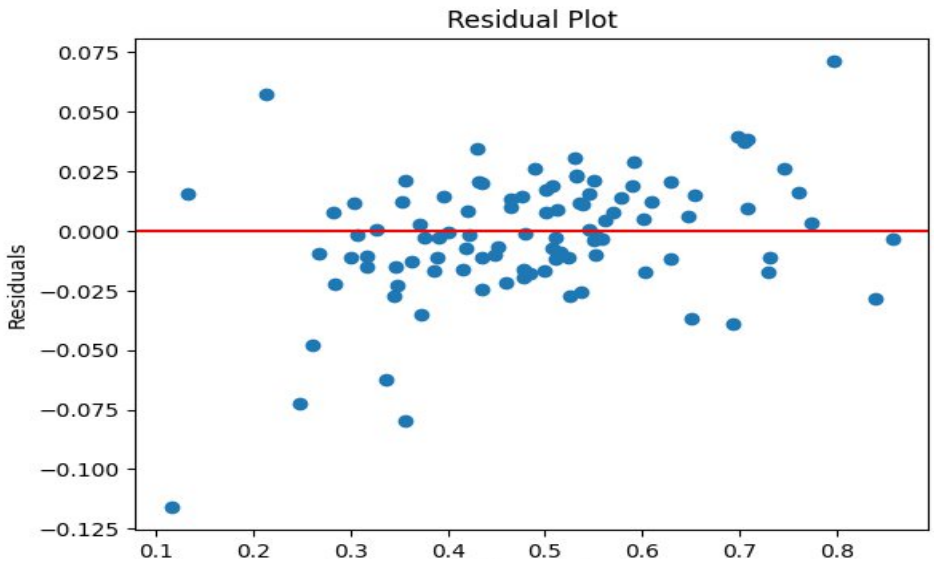


Figure : Residual plot of the GBDT model.

This finding reinforces the robustness and reliability of the model, aligning with the previously noted high R^2 score of 0.973 and the low RMSE of 0.0262. Collectively, these results affirm that the model is highly accurate in predicting car sales, making it a valuable tool for the company's strategic decision-making and sales optimization efforts.

6. Conclusion

The thesis concludes that machine learning, particularly GBDT, offers significant advantages in generating accurate sales forecasts, which can aid businesses in optimizing their operations, reducing waste, and increasing profit

This thesis is outlines the transformative impact of machine learning across various sectors, including sales and marketing. It highlights the limitations of traditional sales forecasting methods, which often lack insights into consumer purchasing patterns, making it difficult for companies to remain competitive. The thesis aims to develop a predictive model for a car seller company by leveraging historical sales data through the Gradient Boosting Decision Tree Algorithm (GBDTA).

The research involves data collection, preprocessing, feature selection, and model training, with performance evaluated using metrics such as Mean Squared Error (MSE) and R^2 score to assess accuracy and effectiveness. The key contributions of this thesis lie in its application of machine learning to predict future sales, thereby providing valuable insights for businesses to enhance their operations and strategic planning.

Future Work

In Ethiopia, numerous companies, industries, and large supermarkets continue to operate using traditional sales methods, which often lack reliable forecasts for future sales. This research highlights the importance of sales prediction as a means to establish concrete goals and benchmarks in the present. The objective of forecasting extends beyond merely predicting future outcomes; it aims to provide actionable insights that can inform decision-making in the present. Moving forward, we plan to implement sales prediction using machine learning algorithms in one of the companies or industries in Ethiopia, thereby enhancing their operational strategies and competitiveness in the market.

REFERENCES

[1] Z. Shilong, Machine learning model for sales forecasting by using XGBoost, 2021 IEEE international conference on consumer electronics and computer engineering (ICCECE), IEEE, 2021, pp. 480-483. describe the actual vs predicted sales graph from the thesis

[2] Y. Niu, Walmart sales forecasting using xgboost algorithm and feature engineering, 2020 International Conference on Big Data & Artificial Intelligence & Software Engineering (ICBASE), IEEE, 2020, pp. 458-461.

[3] S. Cheriyan, S. Ibrahim, S. Mohanan, S. Treesa, Intelligent sales prediction using machine learning techniques, 2018 International Conference on Computing, Electronics & Communications Engineering (iCCECE), IEEE, 2018, pp. 53-58.

[4] P. Bajaj, R. Ray, S. Shedge, S. Vidhate, N. Shardoor, Sales prediction using machine learning algorithms, International Research Journal of Engineering and Technology (IRJET), 7 (2020) 3619-3625.

[5] P. Chate, S.N. Supe, N.H. Tadalapure, S.N. Ovhal, S.G. Wabale, Analogous Study and Performance evaluation of Supervised Machine Learning algorithms for Sales Prediction.

[6] G. Behera, N. Nain, Grid search optimization (GSO) based future sales prediction for big mart, 2019 15th International Conference on Signal-Image Technology & Internet-Based Systems (SITIS), IEEE, 2019, pp. 172-178.

[7] C.-S.M. Wu, P. Patil, S. Gunaseelan, Comparison of different machine learning algorithms for multiple regression on black friday sales data, 2018 IEEE 9th International Conference on Software Engineering and Service Science (ICSESS), IEEE, 2018, pp. 16-20.

[8] M. Singh, B. Ghutla, R.L. Jnr, A.F. Mohammed, M.A. Rashid, Walmart's Sales Data Analysis-A Big Data Analytics Perspective, 2017 4th Asia-Pacific World Congress on Computer Science and Engineering (APWC on CSE), IEEE, 2017, pp. 114-119.

[9] S. Marsland, Machine learning: an algorithmic perspective, Chapman and Hall/CRC2011.

[10] E. Alpaydin, Introduction to machine learning, MIT press2020.

[11] A.W. Tsui, Y.-H. Chuang, H.-H. Chu, Unsupervised learning for solving RSS hardware variance problem in WiFi localization, Mobile Networks and Applications, 14 (2009) 677-691.

[12] J.H. Friedman, Greedy function approximation: a gradient boosting machine, Annals of statistics, (2001) 1189-1232.

[13] T. Chen, C. Guestrin, Xgboost: A scalable tree boosting system, Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining, 2016, pp. 785-794.